

FCFS Appointment Simulation: Readable Sensitivity Report

CUIMC Appointment Simulation

May 7, 2026

1 Executive Summary

This report studies the first policy in the simulation: first-come, first-served (FCFS). The goal is to understand which parameters change total performance and which parameters benefit one class over the other.

Question	Short answer
What matters most overall?	Total demand and no-show behavior. Higher demand lowers served rate and raises offered wait. Higher no-show risk wastes booked slots.
What creates class benefit?	Differences between the classes. The class with lower cancellation, lower balking, higher balking threshold, lower no-show risk, or higher no-show threshold tends to benefit.
Why not use utilization alone?	A busy clinic can still have poor access. In the baseline, utilization is high, but only about 27% of arrivals complete service.
How should plots be read?	The line plots show simple one-at-a-time changes. The heatmaps show what happens when Class 1 and Class 2 parameters vary together.

2 Metrics and Baseline

The analysis uses rates and averages, not total counts. This keeps scenarios comparable when arrival volume changes.

Plain name	Code name	Meaning
Served rate, Class i	<code>percent_served_i</code>	Share of Class i arrivals that complete service.
Overall served rate	<code>overall_percent_served</code>	Share of all arrivals that complete service.
Utilization	<code>average_utilization</code>	Share of daily slots that become completed visits. This is aggregate only.
Offered wait	<code>mean_offered_booking_delay</code>	Average delay offered to patients who received an offer, including patients who balked.
Class gap	<code>percent_served_1 - percent_served_2</code>	Positive means Class 1 has higher served rate. Negative means Class 2 has higher served rate.

Baseline uses `configs/baseline.yaml`: $\lambda_1 = 50$, $\lambda_2 = 50$, 32 slots per day, 14-day horizon, balking threshold 9, balking high probability 0.50, no-show threshold 6, no-show high probability 0.30, and cancellation probability 0.10 for both classes.

Scenario	Utilization	Overall served	Accepted wait	Offered wait	Class gap	Delay gap
Baseline	0.839	0.269	8.35	9.30	0.001	0.003
Scenario 2	1.000	0.395	4.29	5.06	-0.008	0.005

Table 1: Scenario values are rates or averages.

3 How To Read The Figures

Each behavior section has the same visual logic.

- The slice plots are the easiest to read. One class changes while the other class stays at the baseline value.
- The interaction heatmap is for the two-variable question. It shows what happens when Class 1 and Class 2 values move together.
- Blue lines refer to Class 1, orange lines to Class 2, and black lines to the overall value.
- In class-gap heatmaps, blue means Class 1 has the higher served rate and orange means Class 2 has the higher served rate.

4 Balking Step

Balking step is the high-delay rejection probability after the balking threshold. A higher step means patients reject long-delay offers more often.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Class 1 and Class 2 balking step, from 0.00 to 1.00
Fixed at baseline	Balking threshold = 9 days; no-show, cancellation, arrivals, capacity, horizon
Grid and seeds	21 values per class, step size 0.05; single fine-grid seed 2027

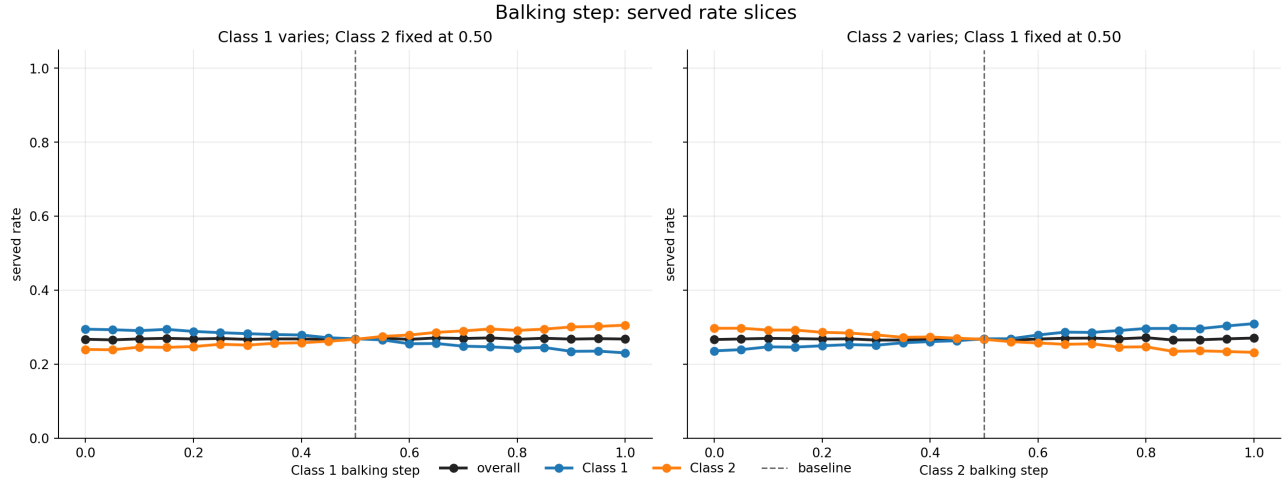


Figure 1: Served-rate slices for balking step.

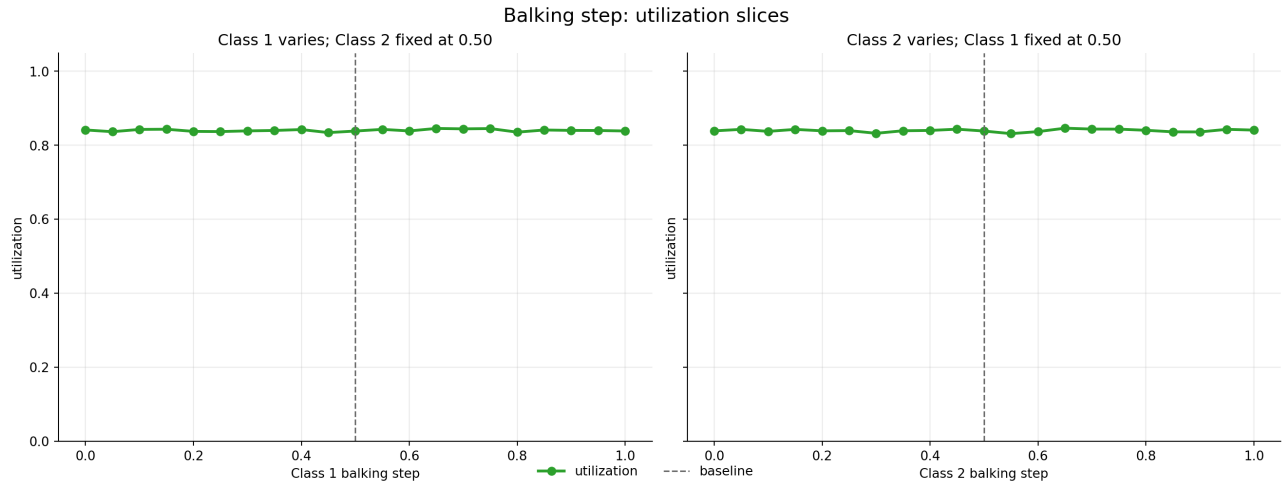


Figure 2: Utilization slices for balking step.

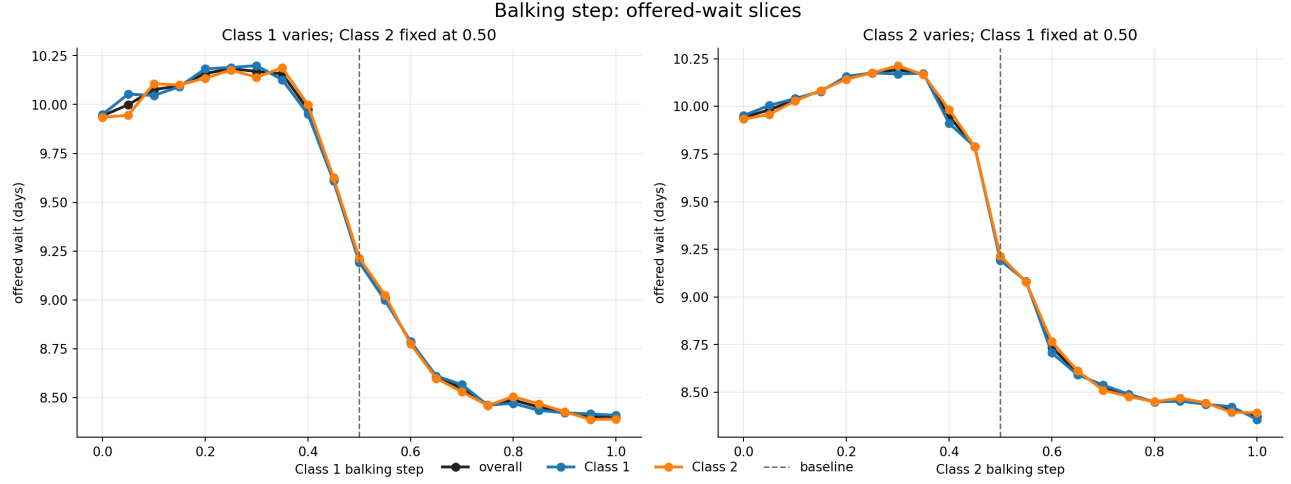


Figure 3: Offered-wait slices for balking step.

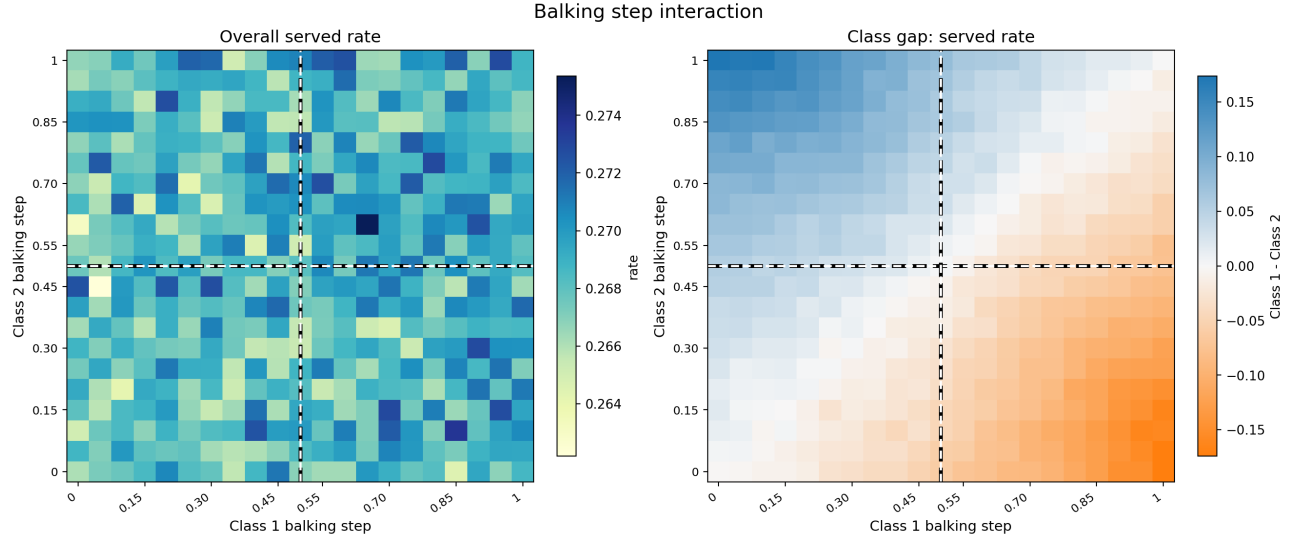


Figure 4: Two-variable balking-step heatmap. Left: overall served rate. Right: class gap.

Main reading. Higher balking step hurts the class whose step increases because more of that class rejects long-delay offers. The other class can gain access to capacity that was released. Utilization moves much less than class served rates because total demand is still high.

Takeaway. Balking step is mostly redistribution. The class that balks less is more likely to be served.

5 Balking Threshold

Balking threshold is the delay where high balking begins. A lower threshold means patients start rejecting earlier. A higher threshold means patients tolerate longer waits.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Class 1 and Class 2 balking threshold, from 0 to 13 days
Fixed at baseline	Balking step = 0.50; no-show, cancellation, arrivals, capacity, horizon
Grid and seeds	14 integer threshold values per class; single fine-grid seed 2027

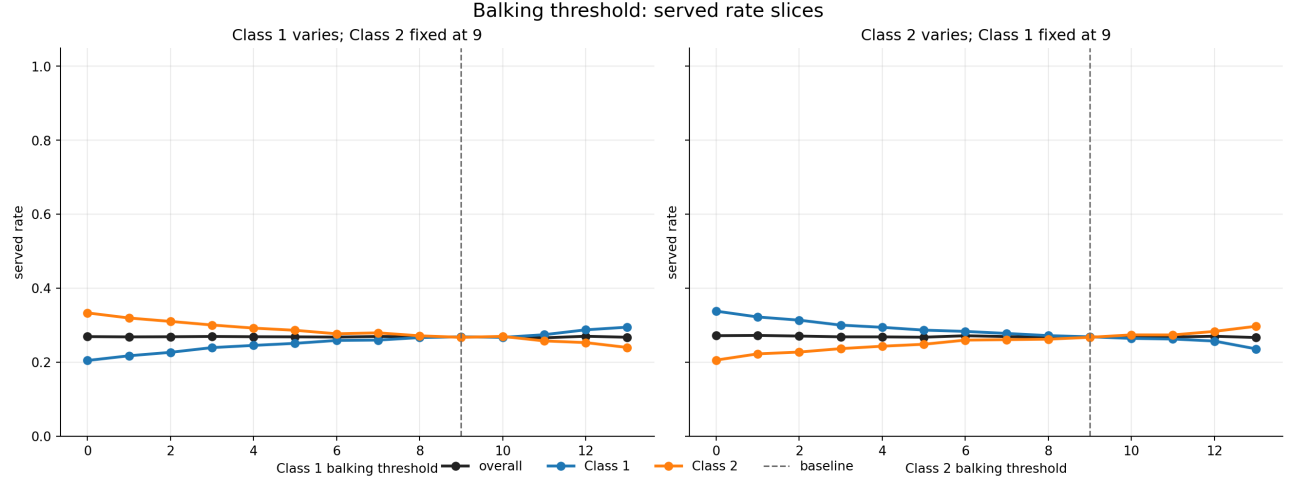


Figure 5: Served-rate slices for balking threshold.

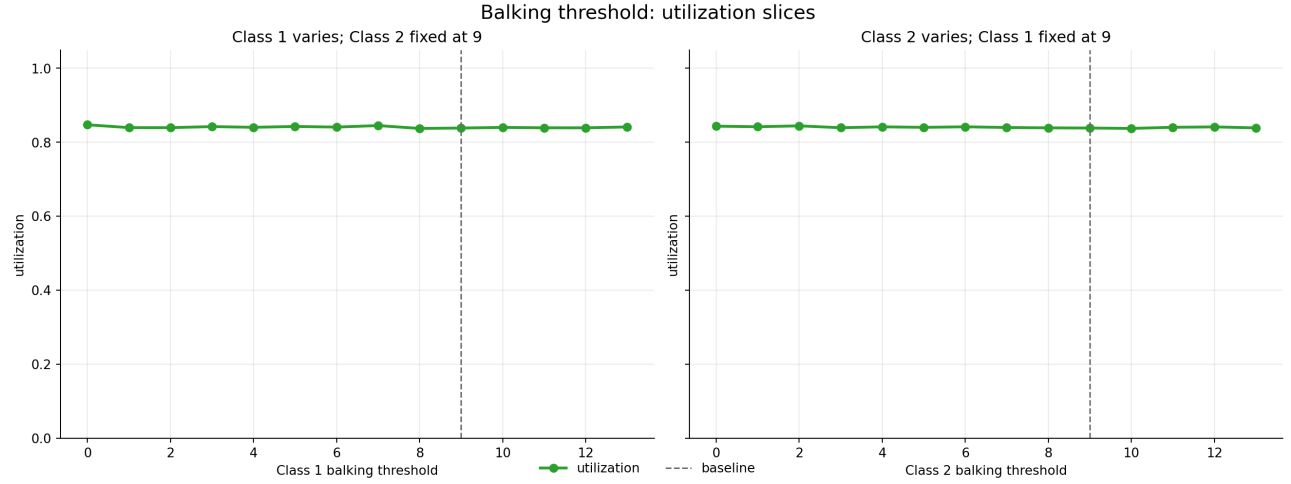


Figure 6: Utilization slices for balking threshold.

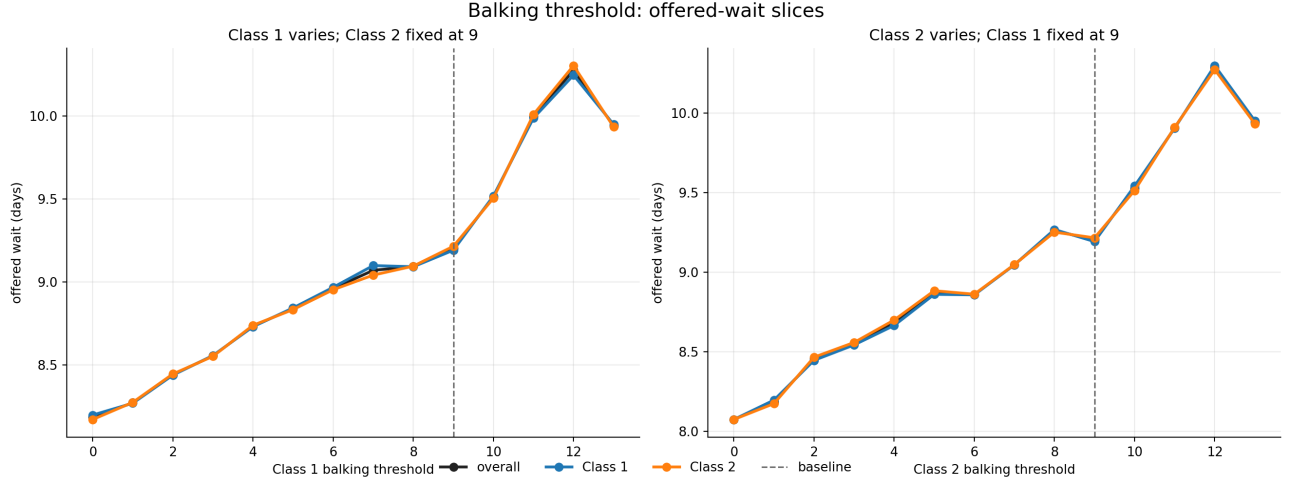


Figure 7: Offered-wait slices for balking threshold.

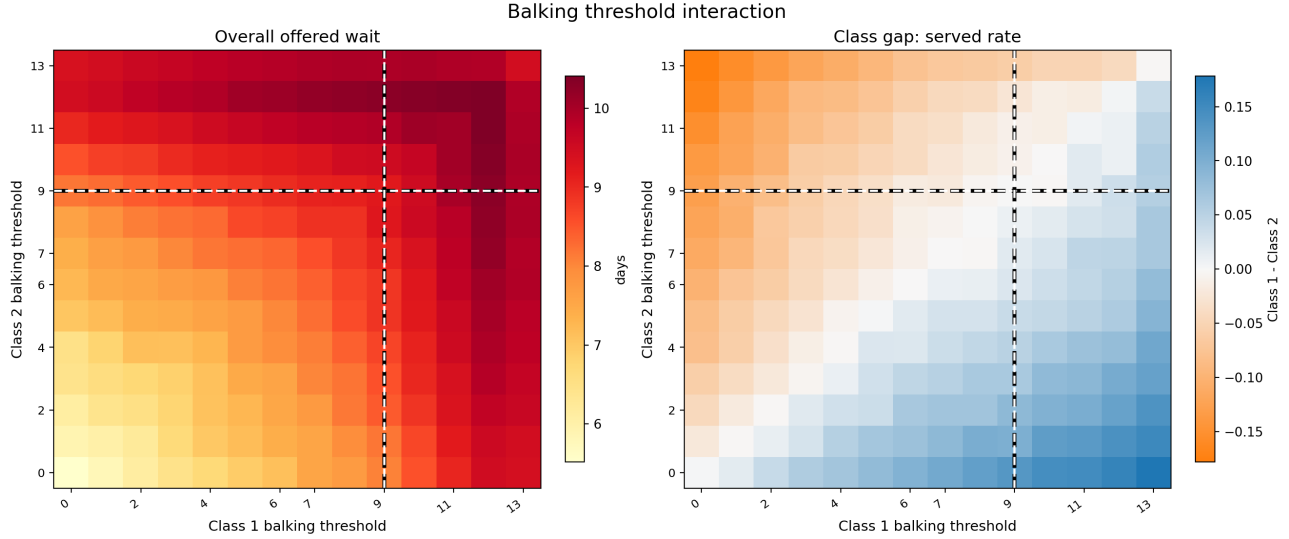


Figure 8: Two-variable balking-threshold heatmap. Left: offered wait. Right: class gap.

Main reading. The class with the higher threshold accepts longer waits and is more likely to complete service. Offered wait rises when thresholds are high because more patients stay in the appointment pool for long-delay offers.

Takeaway. Balking threshold is also mostly redistribution. The class that tolerates longer waits gets more service, but higher tolerance can keep the system more congested.

6 No-Show Step

No-show step is the high-delay probability that a booked patient does not attend. This happens after booking, so it mainly affects completed visits and slot use.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Class 1 and Class 2 no-show step, from 0.00 to 1.00
Fixed at baseline	No-show threshold = 6 days; balking, cancellation, arrivals, capacity, horizon
Grid and seeds	21 values per class, step size 0.05; single fine-grid seed 2027

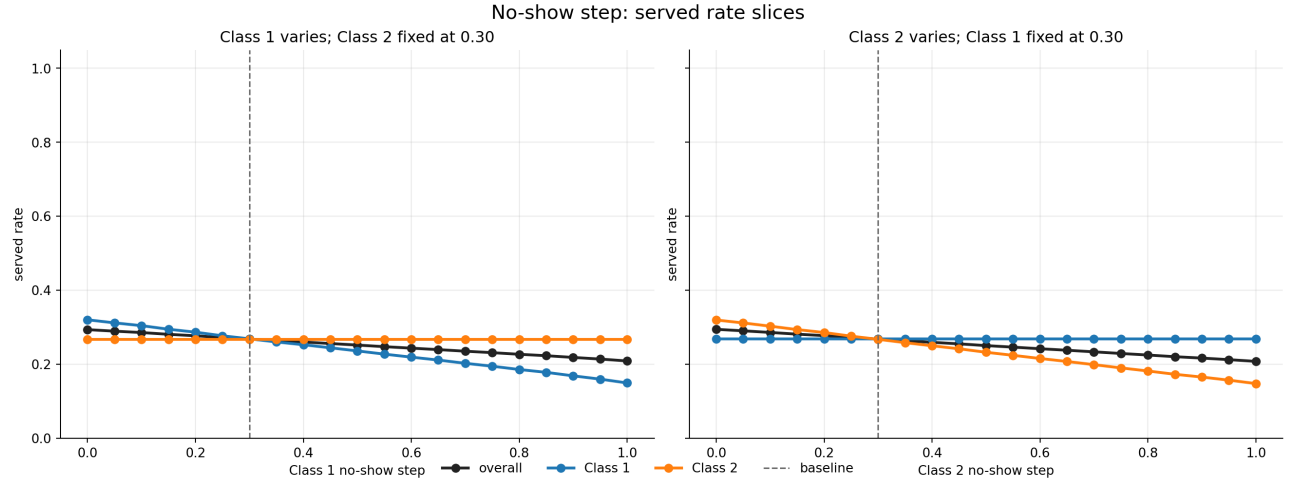


Figure 9: Served-rate slices for no-show step.

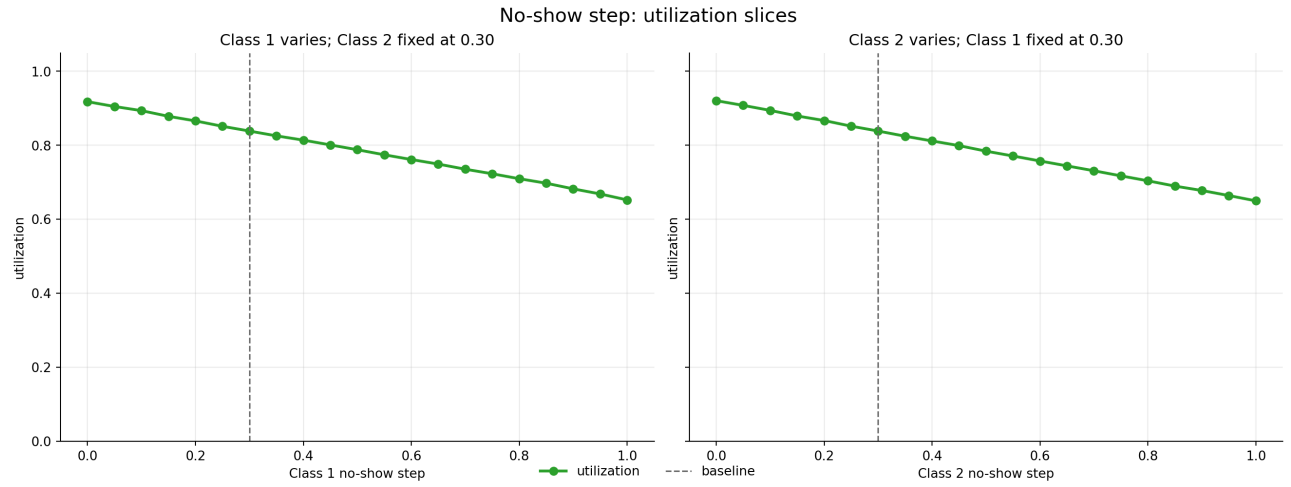


Figure 10: Utilization slices for no-show step.

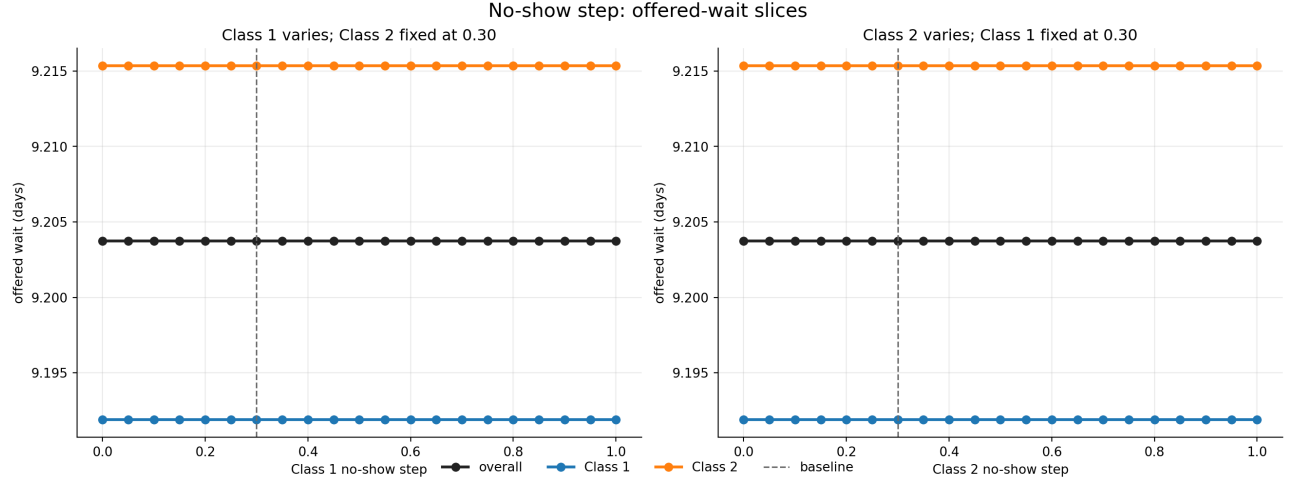


Figure 11: Offered-wait slices for no-show step.

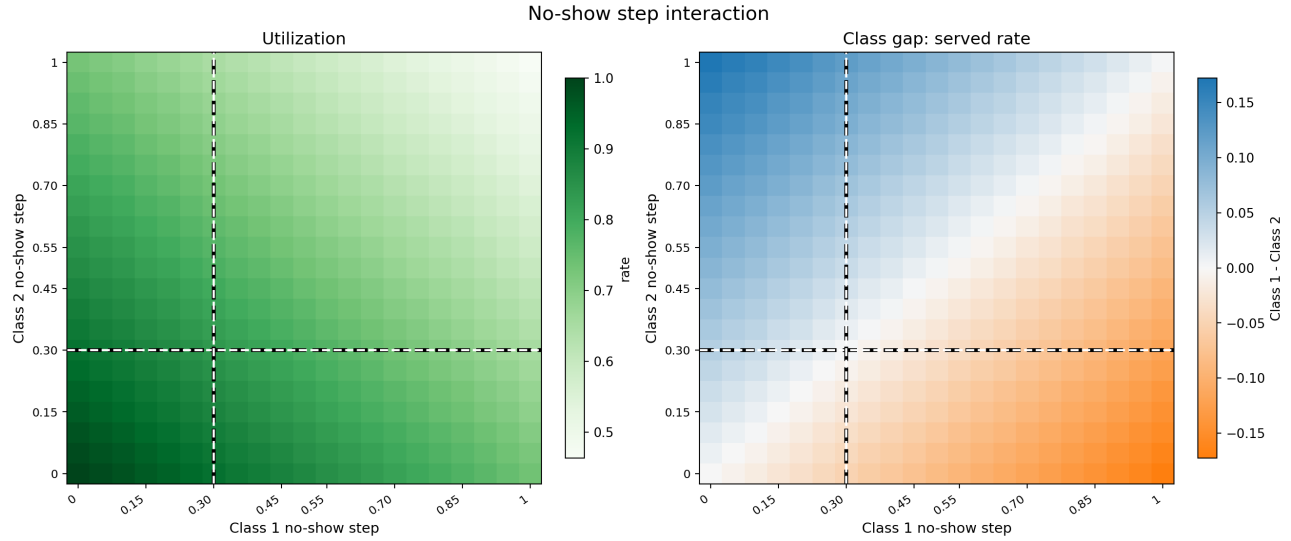


Figure 12: Two-variable no-show-step heatmap. Left: utilization. Right: class gap.

Main reading. Higher no-show step reduces completed visits. The class with higher no-show risk loses served rate after it has already accepted appointments. Offered wait changes little because the wait was assigned before the no-show event.

Takeaway. No-show step is both efficiency and redistribution. It wastes capacity and benefits the class with lower no-show risk.

7 No-Show Threshold

No-show threshold is the delay where high no-show risk begins. A higher threshold delays the point where no-show risk increases.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Class 1 and Class 2 no-show threshold, from 0 to 13 days
Fixed at baseline	No-show step = 0.30; balking, cancellation, arrivals, capacity, horizon
Grid and seeds	14 integer threshold values per class; single fine-grid seed 2027

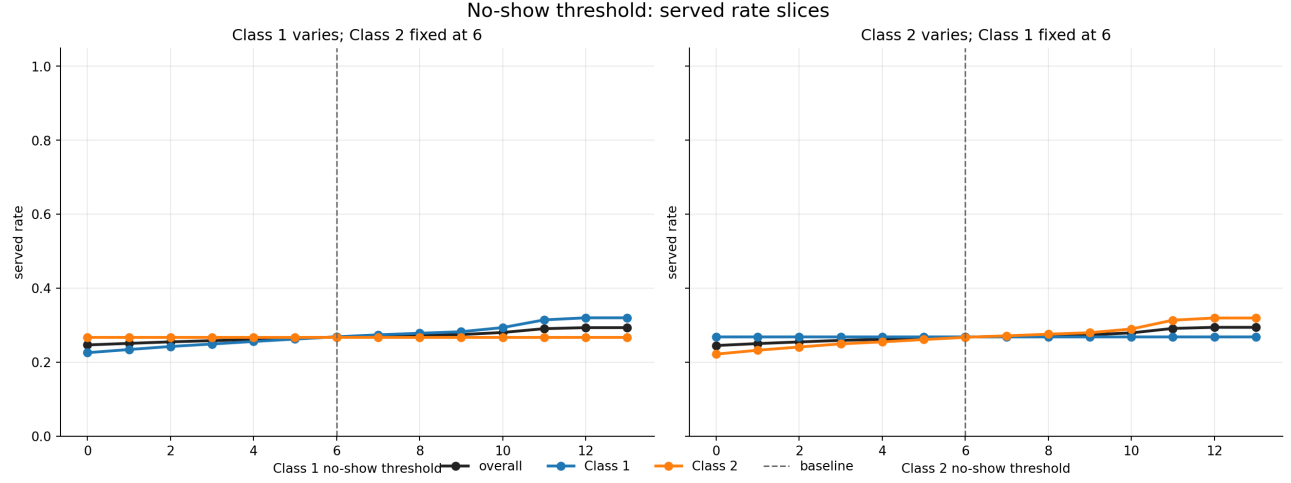


Figure 13: Served-rate slices for no-show threshold.

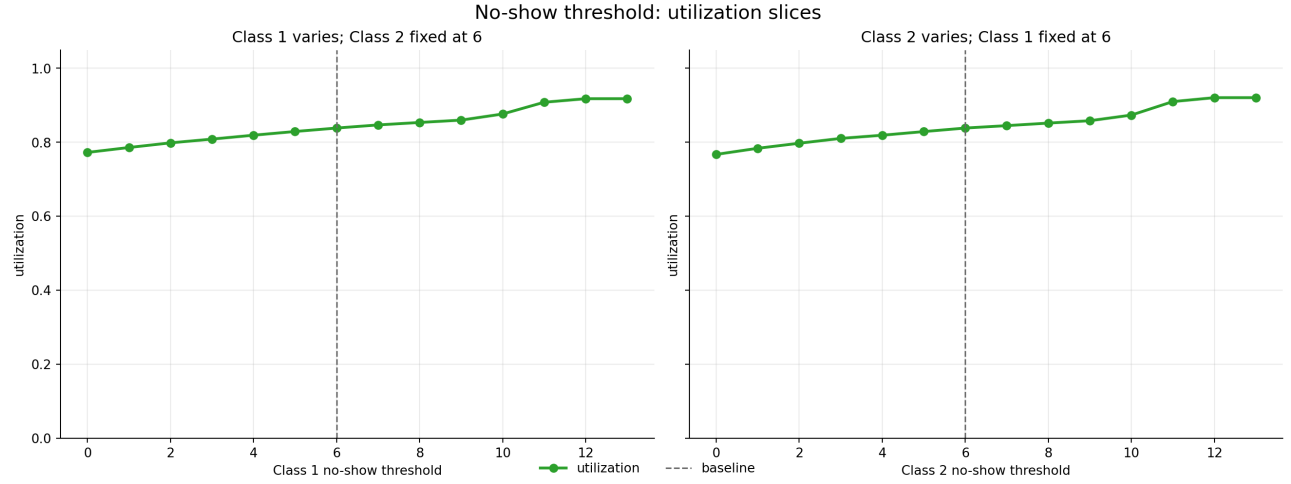


Figure 14: Utilization slices for no-show threshold.

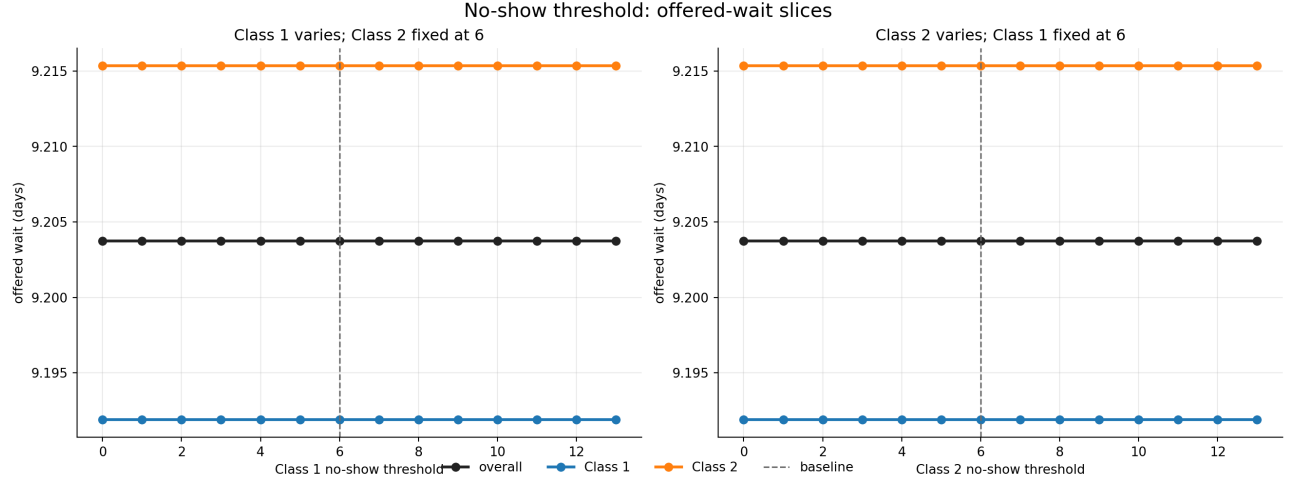


Figure 15: Offered-wait slices for no-show threshold.

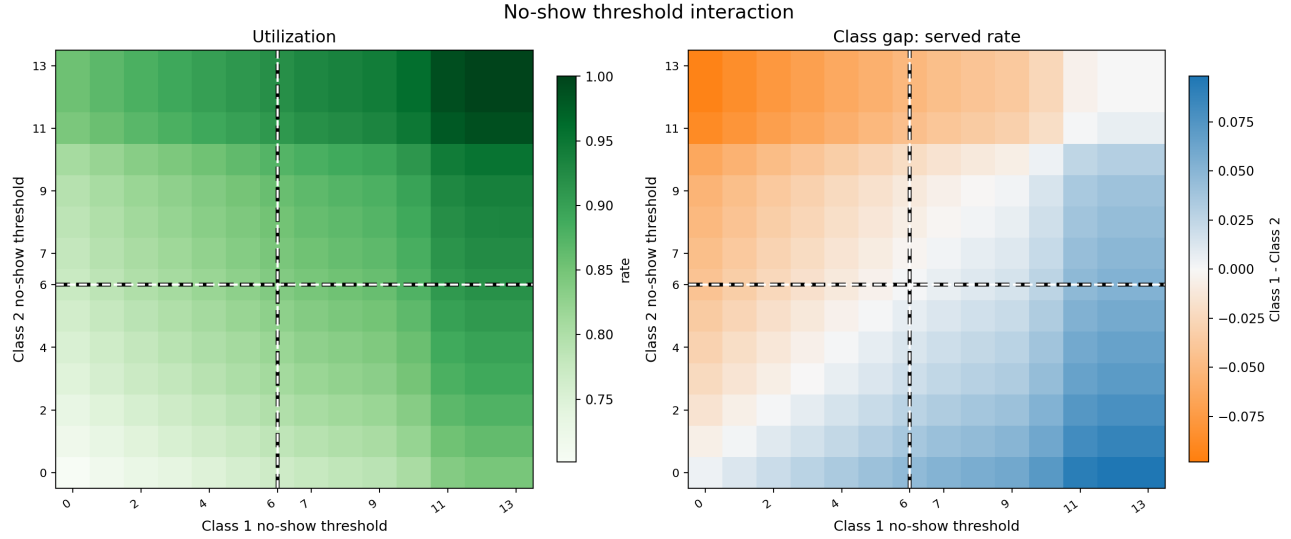


Figure 16: Two-variable no-show-threshold heatmap. Left: utilization. Right: class gap.

Main reading. A higher no-show threshold helps because patients become high-risk only at longer delays. The class with the higher threshold completes more booked appointments. Again, offered wait barely moves compared with served rate and utilization.

Takeaway. No-show threshold is mainly an efficiency parameter when both classes move together, and a redistribution parameter when thresholds differ by class.

8 Cancellation Probability

Cancellation is a direct probability that a booked appointment is canceled before the service day.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Class 1 and Class 2 cancellation probability, from 0.00 to 0.30
Fixed at baseline	Balking, no-show, arrivals, capacity, horizon
Grid and seeds	21 values per class, step size 0.015; single fine-grid seed 2027

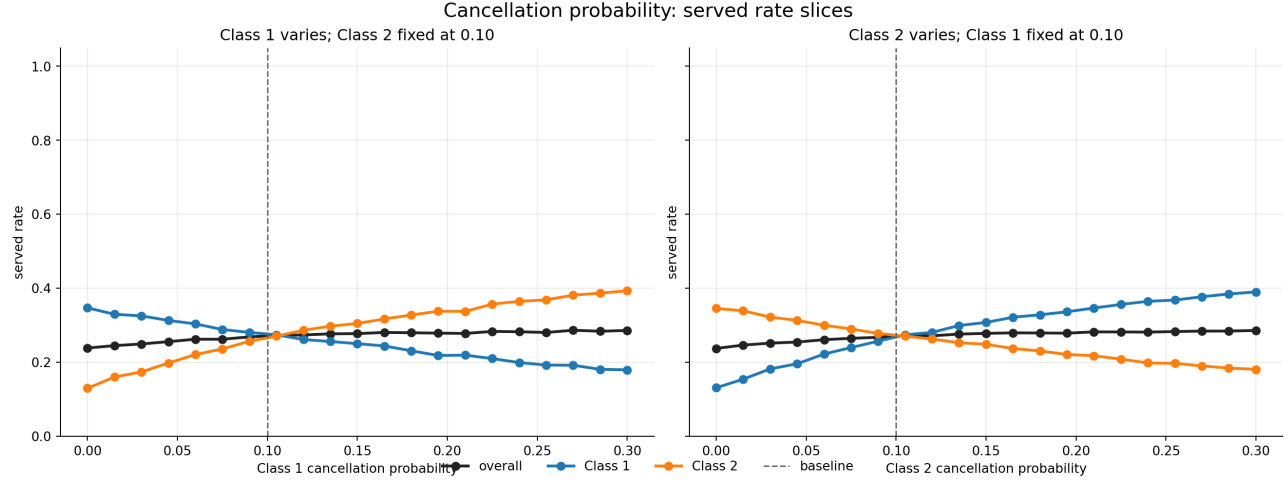


Figure 17: Served-rate slices for cancellation probability.

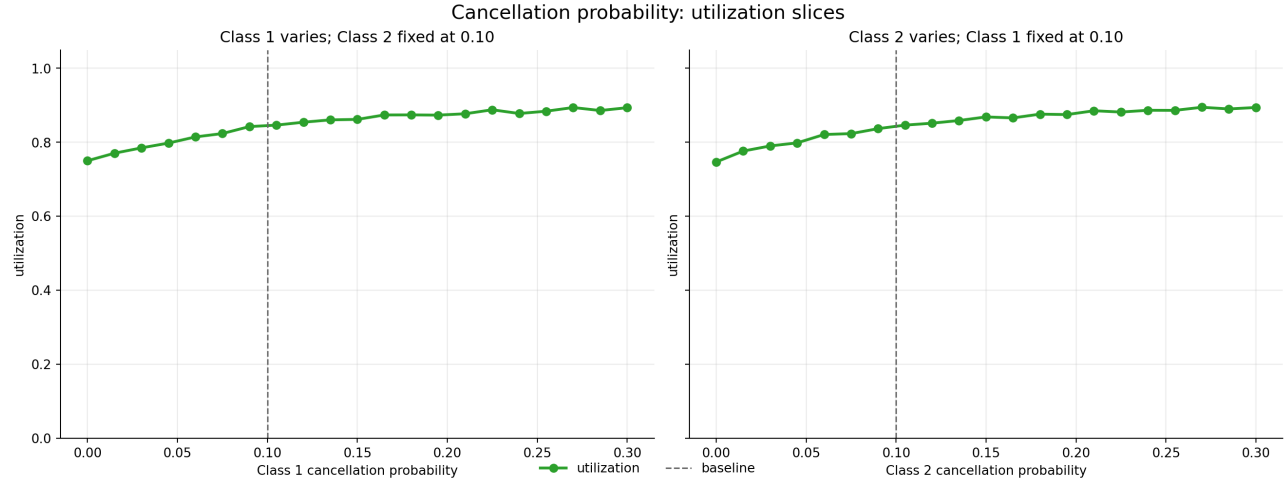


Figure 18: Utilization slices for cancellation probability.

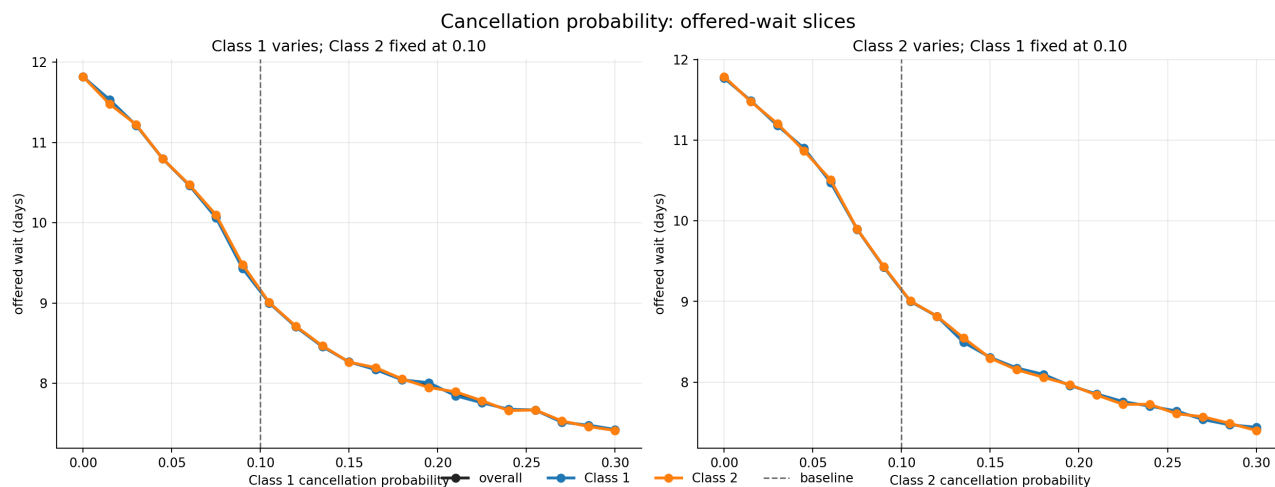


Figure 19: Offered-wait slices for cancellation probability.

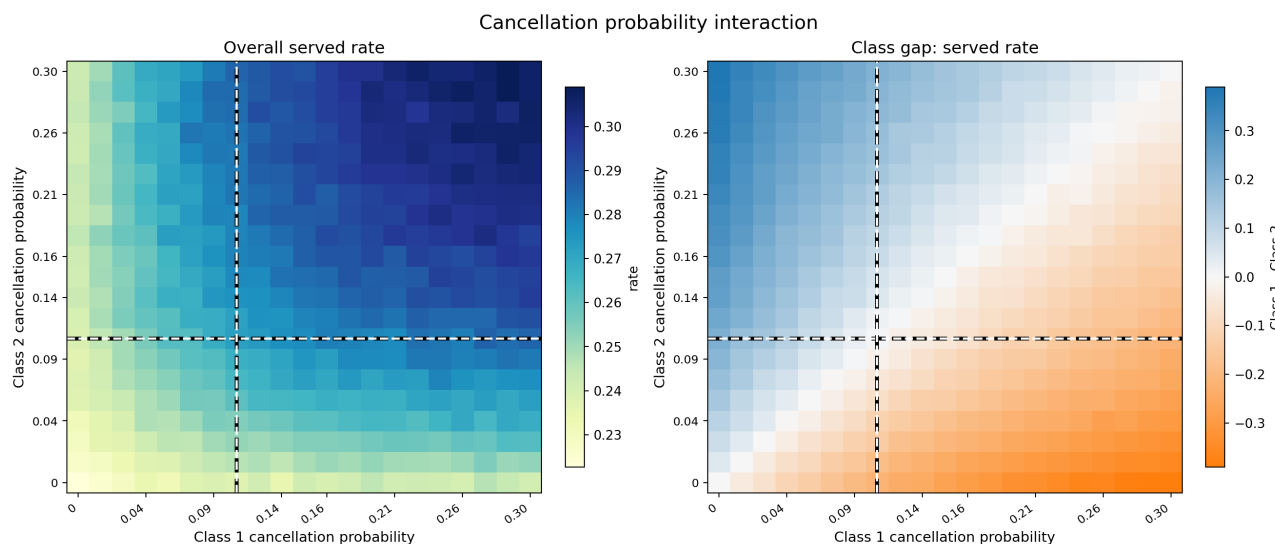


Figure 20: Two-variable cancellation heatmap. Left: overall served rate. Right: class gap.

Main reading. The class with higher cancellation probability loses completed visits. Canceled appointments may free capacity later, but the canceling patient is still not served.

Takeaway. Cancellation is one of the clearest class-gap drivers. The class with lower cancellation probability is much more likely to complete service.

9 Arrival Rate and Class Mix

Arrival parameters change how much demand enters the system. They do not directly change behavior, but they change congestion.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Total arrivals per day and Class 1 arrival share
Fixed at baseline	Balking, no-show, cancellation, capacity, horizon
Grid and seeds	Total arrivals from 50 to 150 per day; Class 1 share from 0.10 to 0.90; single fine-grid seed 2027

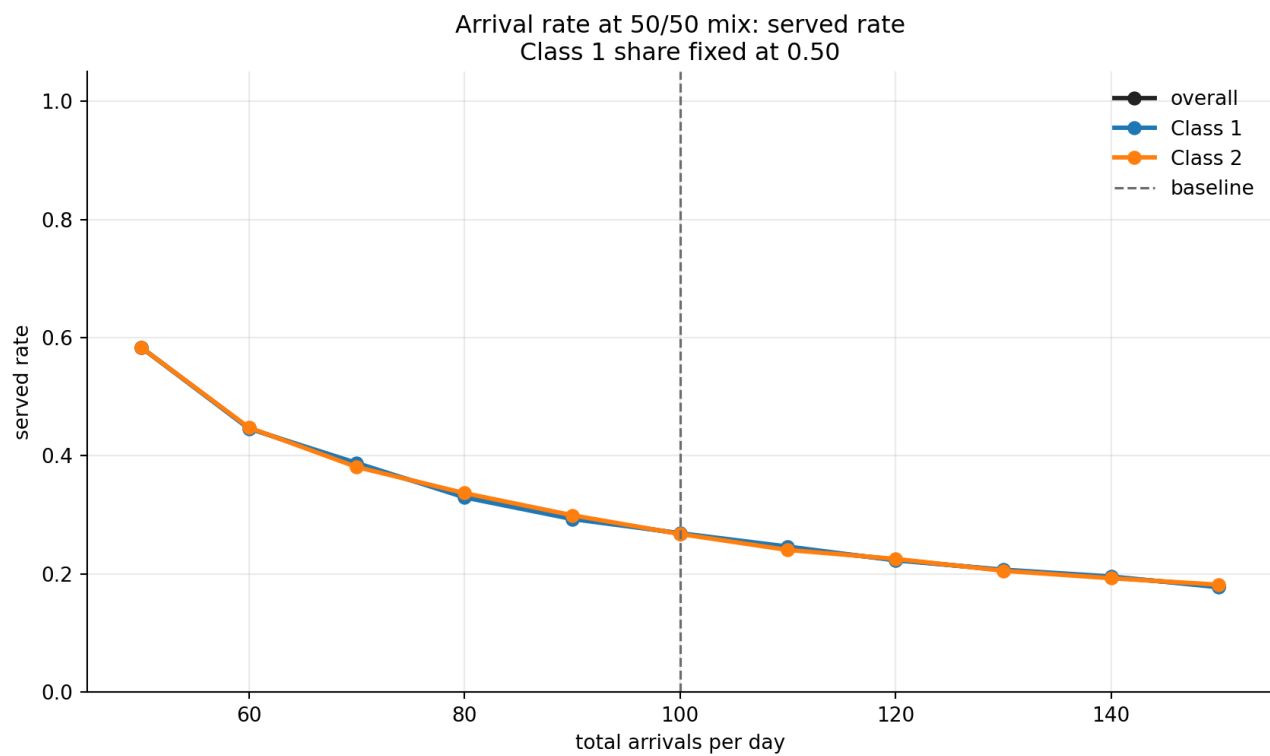


Figure 21: Served-rate slice as total arrivals change and class mix stays 50/50.

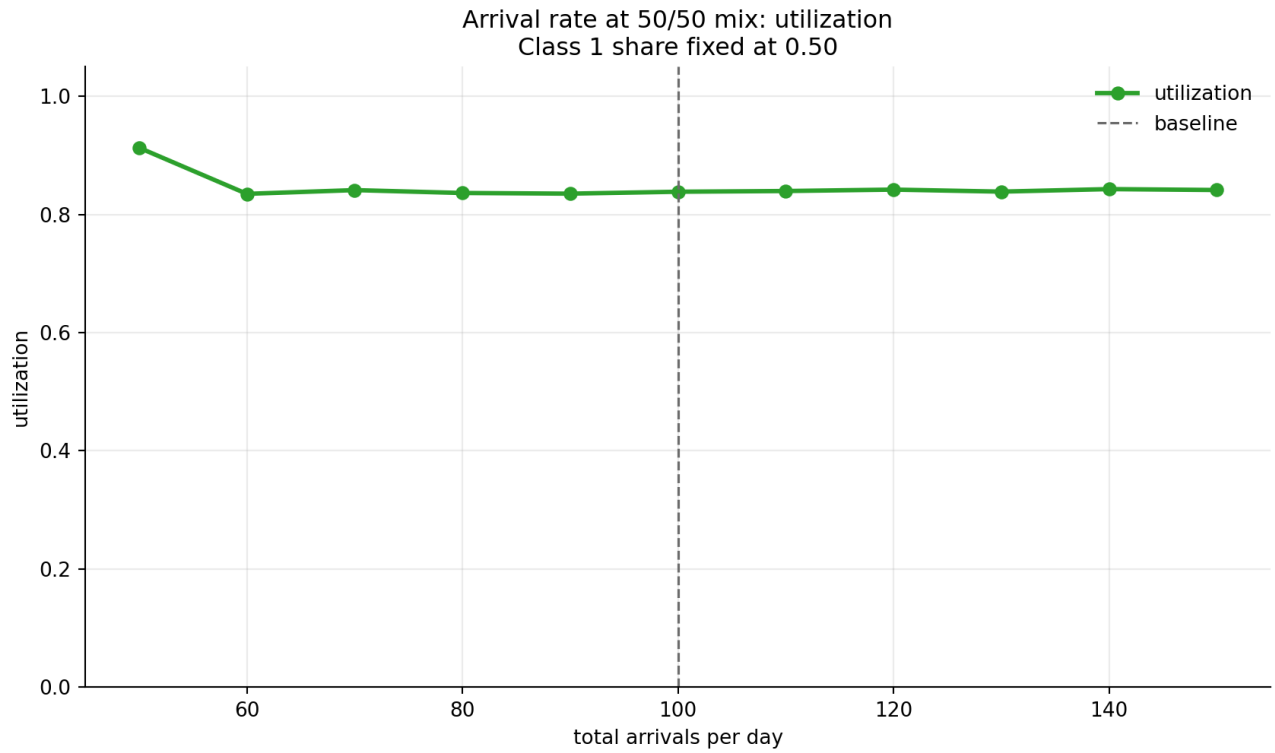


Figure 22: Utilization slice as total arrivals change and class mix stays 50/50.

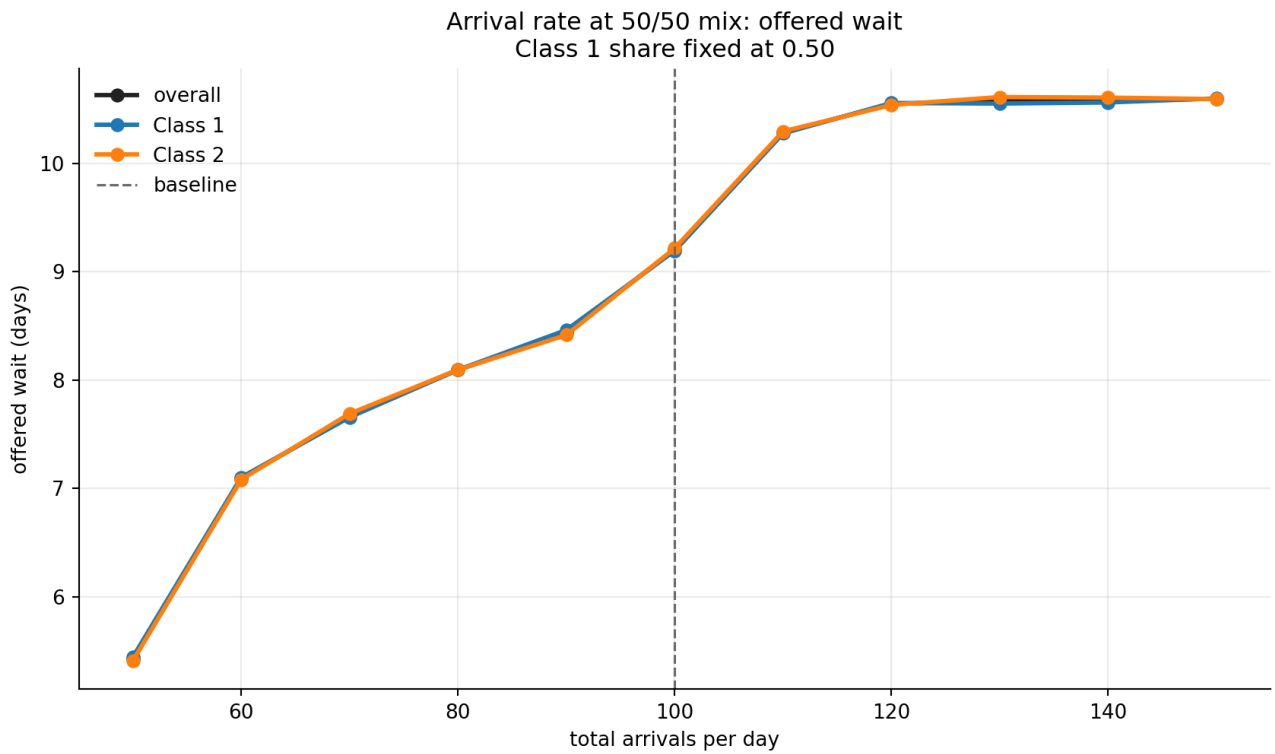


Figure 23: Offered-wait slice as total arrivals change and class mix stays 50/50.

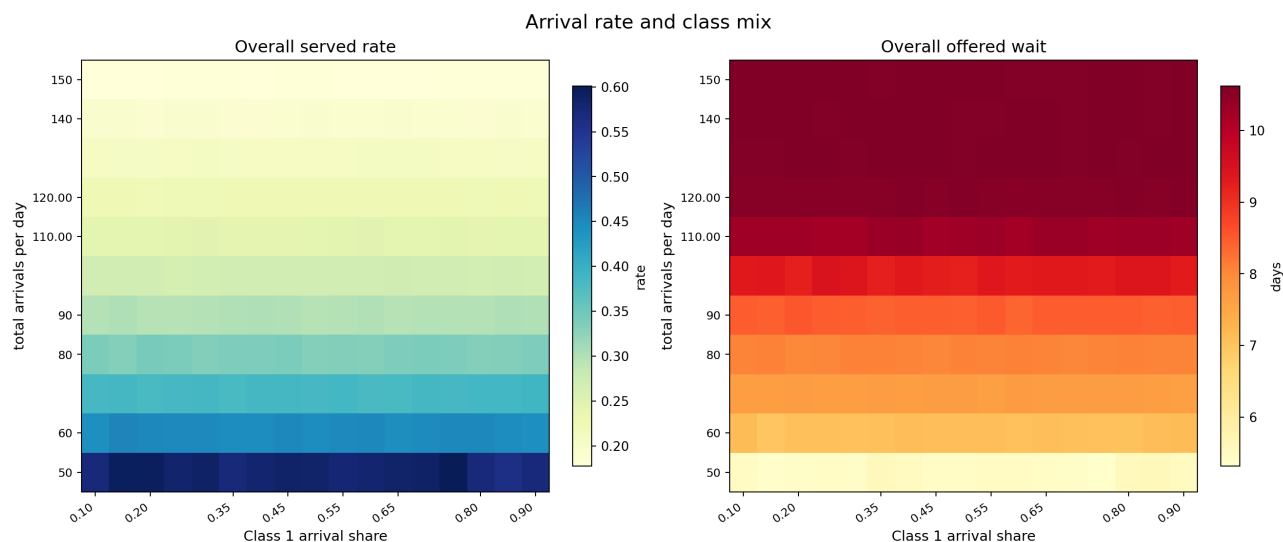


Figure 24: Arrival-rate and class-mix heatmap. Left: overall served rate. Right: offered wait.

Main reading. Higher total demand lowers served rate and raises offered wait. Utilization can remain high even when access is poor. Changing the class mix alone does not create a strong class advantage because the two classes have the same behavior rules.

Takeaway. Arrival rate is mostly an efficiency and congestion parameter. Class mix matters more when the classes also differ in behavior.

10 FCFS Stress Test

The stress test keeps behavior fixed and varies total demand over a wider range. It shows why high utilization is not enough to judge performance.

Policy	FCFS appointment assignment
Baseline demand	Class 1 = 50 arrivals/day; Class 2 = 50 arrivals/day
Capacity and horizon	32 slots/day; 14-day booking horizon; 365 measured days
Varied in this section	Total arrivals per day, preserving the 50/50 class mix
Fixed at baseline	Balking, no-show, cancellation, class mix, capacity, horizon
Grid and seeds	30 to 170 total arrivals per day; seeds 2027, 2028, 2029 averaged

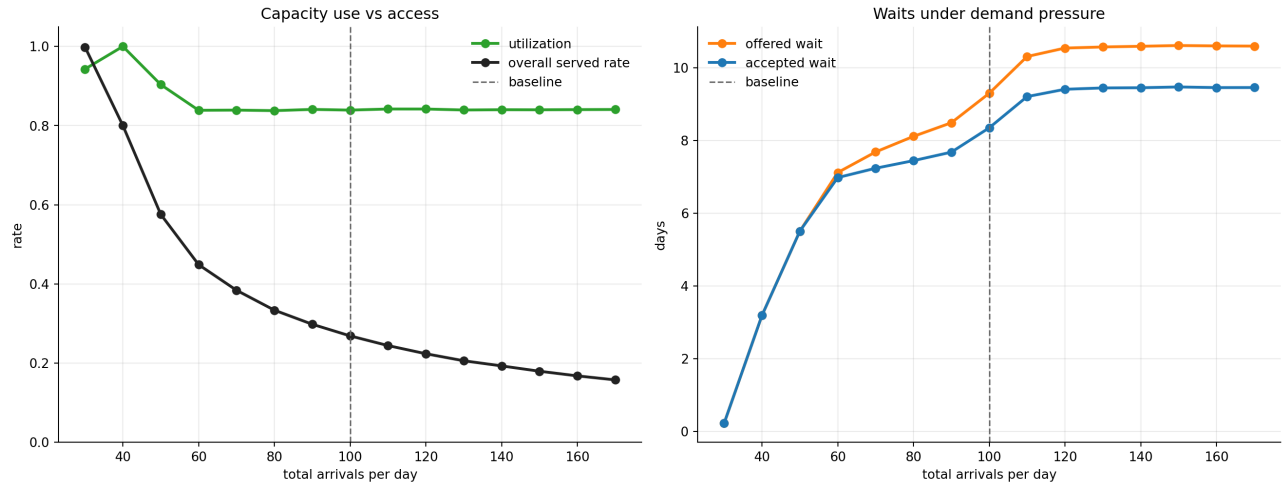


Figure 25: FCFS stress test. Left: utilization versus served rate. Right: accepted and offered wait.

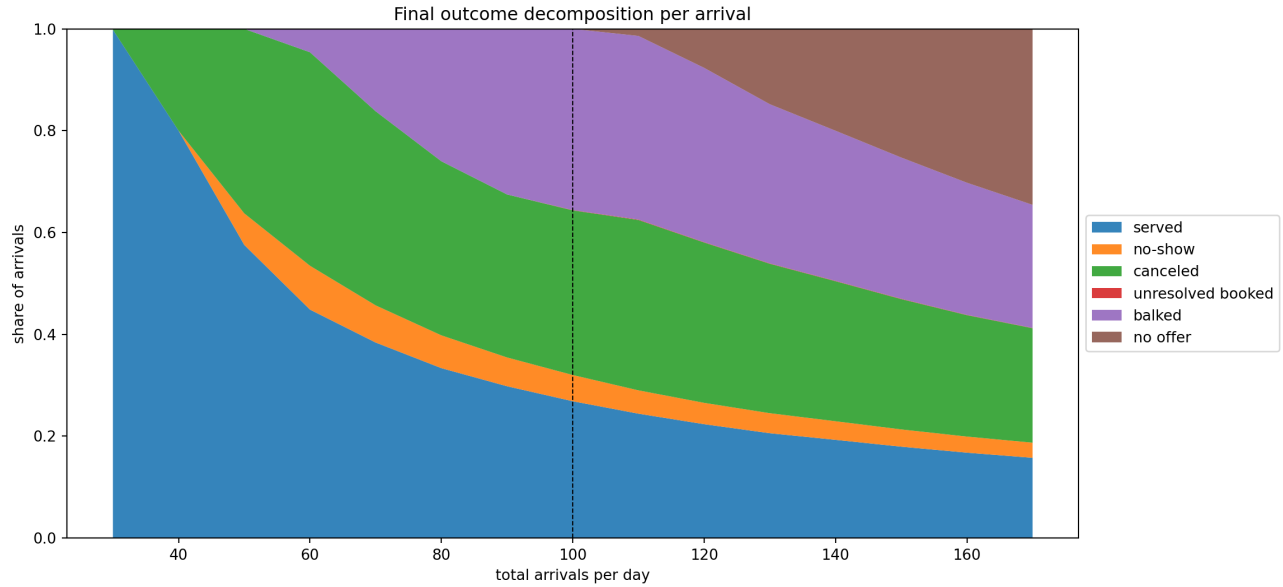


Figure 26: Final outcome decomposition under FCFS. Each band is a share of arrivals.

Arrivals/day	Utilization	Served	Offered wait	Balked	No offer	Lost after booking	Class gap
30	0.943	0.998	0.22	0.000	0.000	0.002	-0.000
100	0.839	0.269	9.30	0.356	0.000	0.376	0.001
170	0.841	0.157	10.59	0.242	0.346	0.255	0.001

Table 2: Selected stress-test points. Outcome columns are shares of arrivals.

Main reading. At low demand, almost every arrival is served. Around baseline demand, utilization is high but served rate is low. At overload, utilization is still high, but no-offers become a major loss channel.

Takeaway. FCFS can look busy while access is poor. Use served rate and outcome decomposition

with utilization.

11 Regression Screening

The heatmaps change one or two parameters at a time. The regression screen asks which inputs explain outcomes across many random FCFS scenarios.

Policy	FCFS appointment assignment
Randomized settings	240 random parameter settings
Seeds	2 seeds per setting, averaged
Inputs	Total demand, class mix, balking steps and thresholds, no-show steps and thresholds, cancellation probabilities
Outputs	Utilization, overall served rate, offered wait, Class 1 served-rate gap

Target	Most important feature	Standardized coefficient
Utilization	Average no-show threshold	0.512
Utilization	Average no-show step	-0.423
Utilization	Average cancellation probability	0.320
Overall served rate	Total arrival rate	-0.774
Overall served rate	Average no-show threshold	0.251
Offered wait	Total arrival rate	0.576
Offered wait	Average cancellation probability	-0.441
Class gap	Cancellation probability gap	-0.452
Class gap	Balking threshold gap	0.429
Class gap	Balking step gap	-0.349

Table 3: Top regression drivers. Gap means Class 1 value minus Class 2 value.

Main reading. Aggregate performance is mostly explained by total demand and average no-show behavior. Class advantage is mostly explained by differences between class behavior parameters.

Takeaway. Absolute demand and no-show levels explain overall performance. Class gaps explain who benefits.

12 Final Takeaways

1. Served rate is the main access metric. Utilization alone can hide poor access.
2. Balking mostly redistributes service toward the class that accepts longer waits.
3. No-show behavior is a strong efficiency driver because booked slots can be wasted.
4. Cancellation differences create large class gaps.
5. Arrival rate controls congestion. Class mix alone is weak when classes behave the same.
6. For class benefit, the difference between Class 1 and Class 2 parameters matters more than the absolute level.

A Appendix: Dense Reference Plots

The main report uses readable slice plots and focused two-panel heatmaps. The figures below are retained for traceability when a fuller diagnostic view is useful.

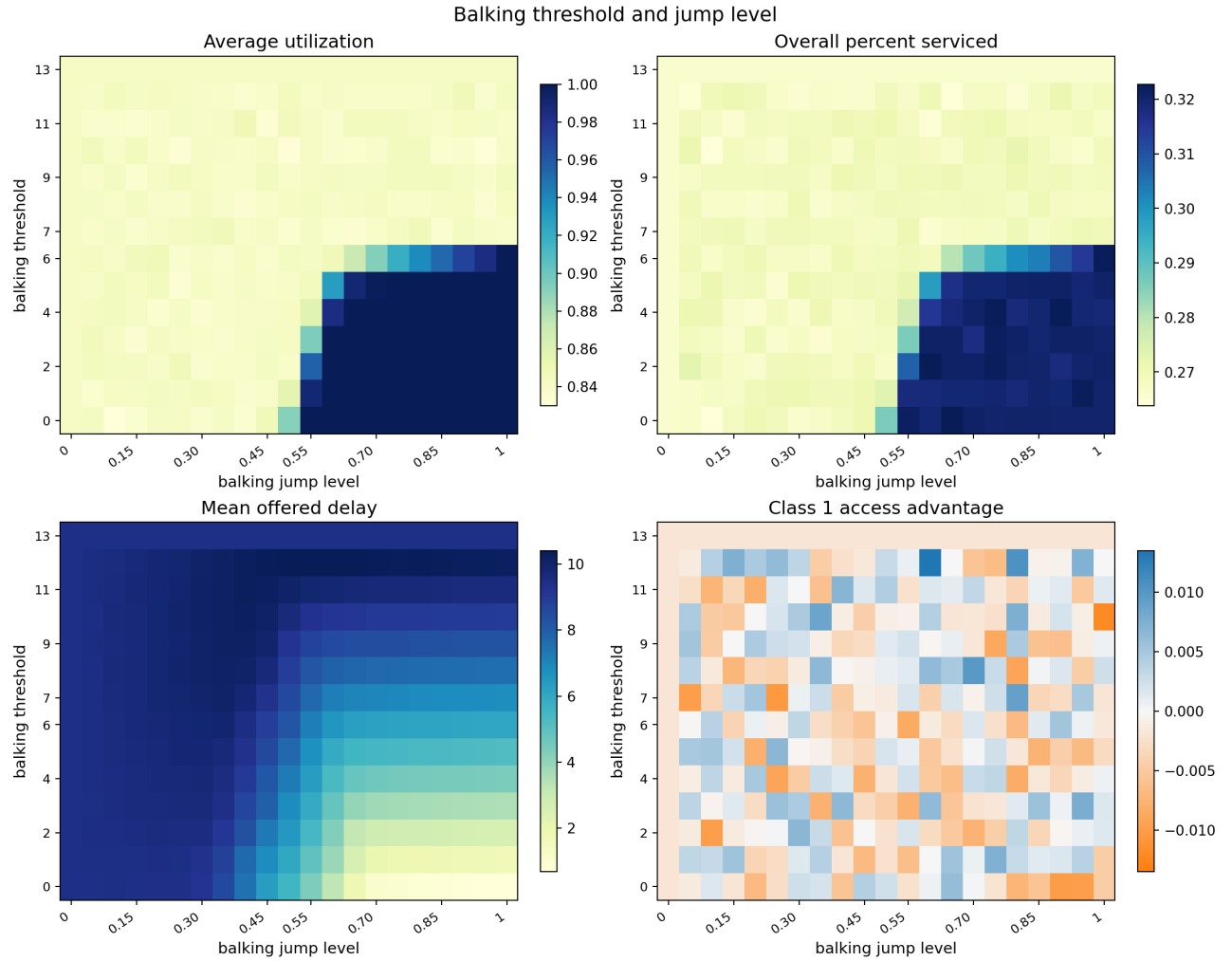


Figure 27: Balking threshold and jump level when both classes share the same values.

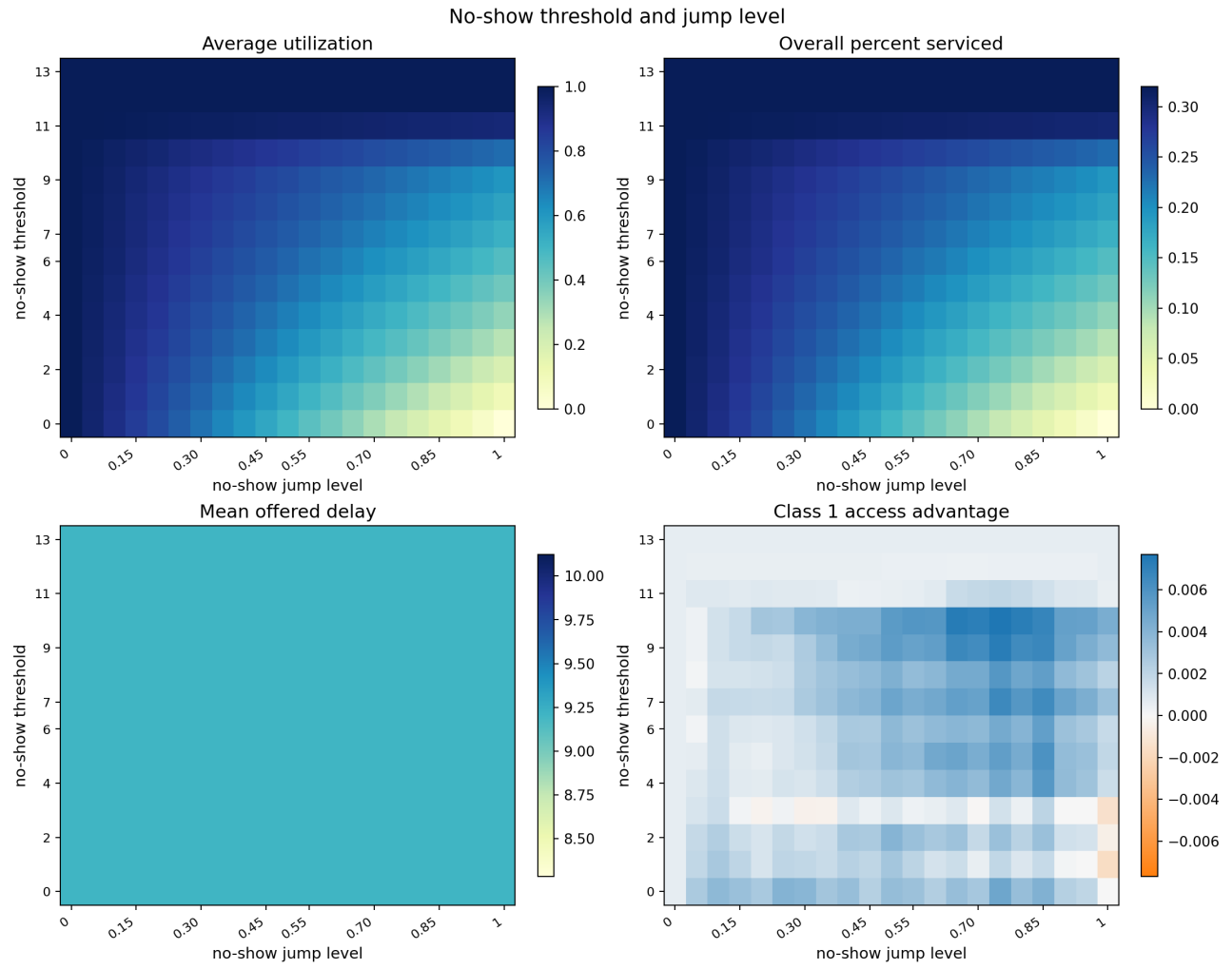


Figure 28: No-show threshold and jump level when both classes share the same values.

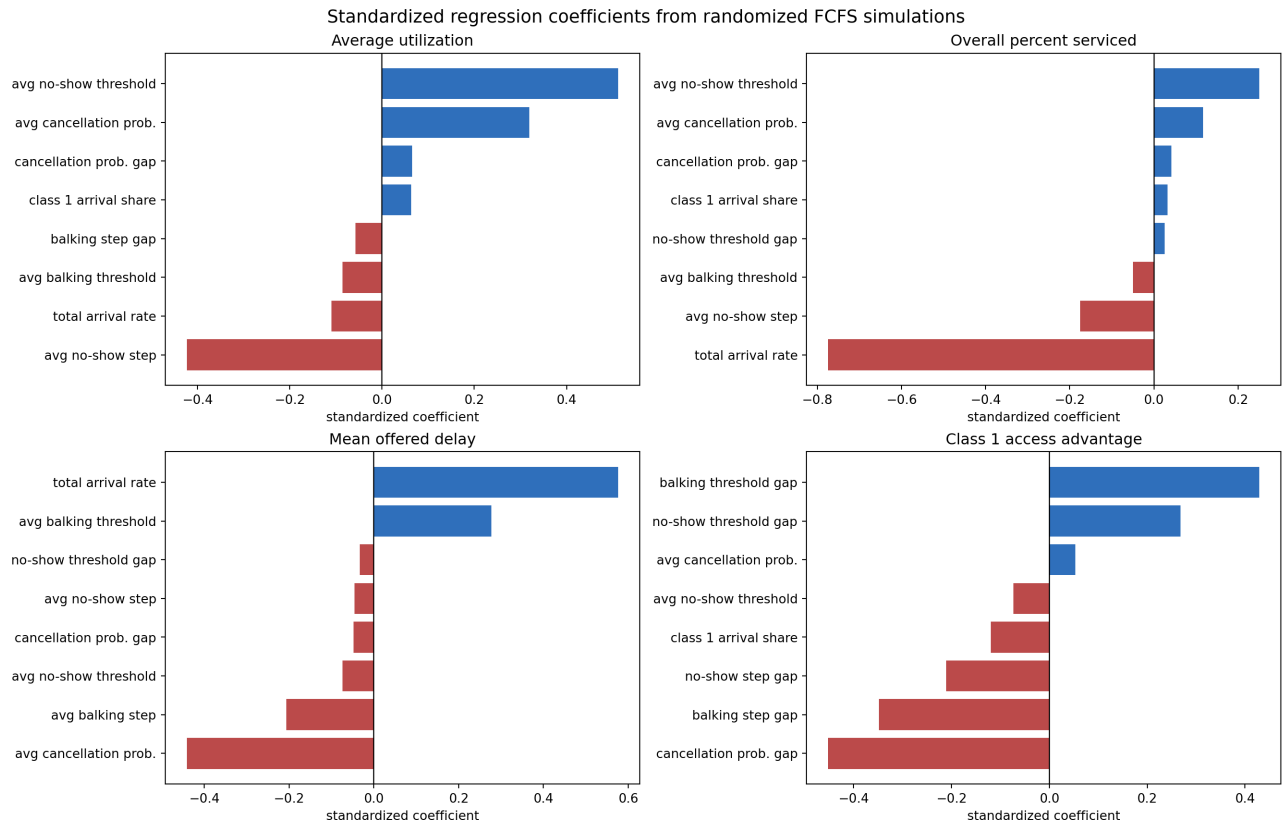


Figure 29: Full standardized regression coefficient plot.

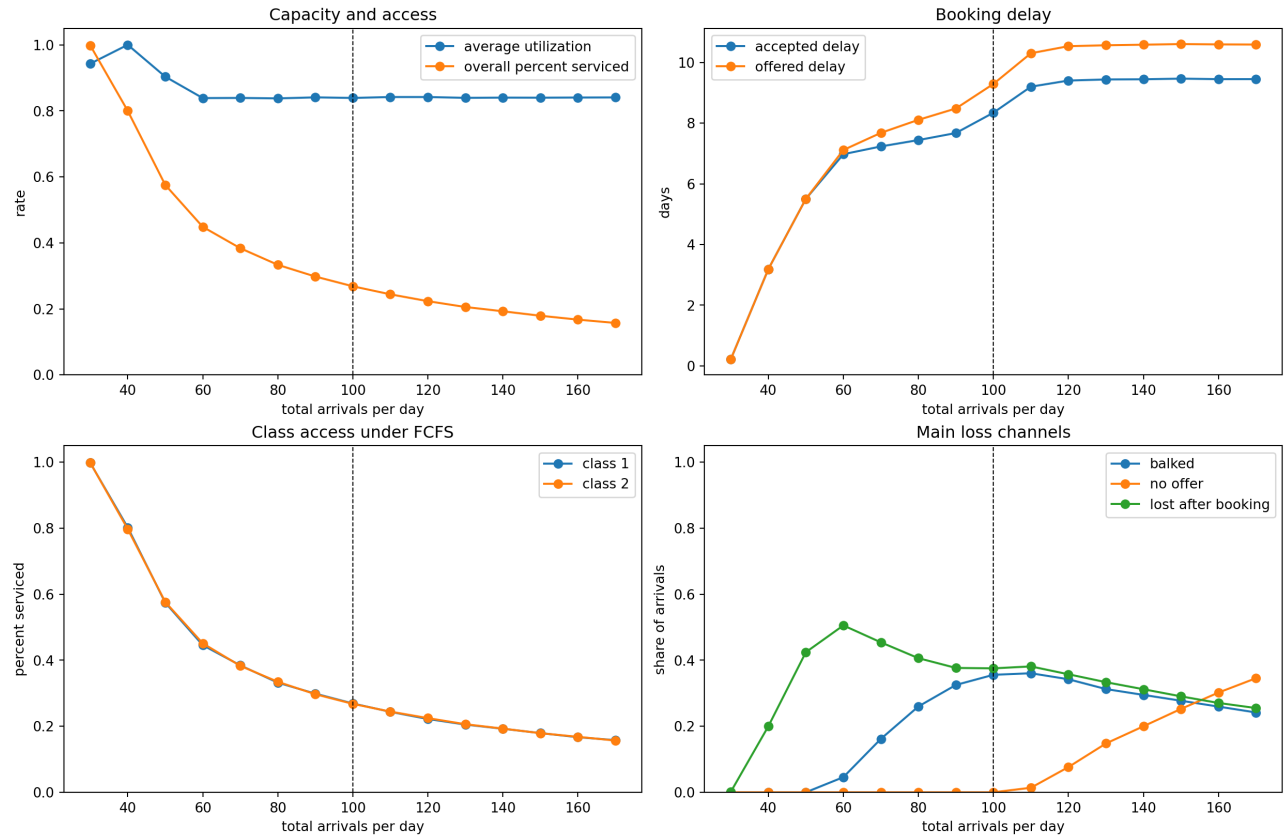


Figure 30: Full FCFS stress diagnostic plot.